

Math 1610, Honors Calculus, Fall 2026,
practice problems for Exam 1

Some definitions you may want.

A **field** is a set F with two operations $+$ and $\cdot$ and two distinguished elements $0 \neq 1$, such that:

- $+$ is associative and commutative, $a + 0 = a$ for all a , and every $a \in F$ has an additive inverse $-a$ with $a + (-a) = 0$;
- $\cdot$ is associative and commutative, $a \cdot 1 = a$ for all a , and every $a \neq 0$ has a multiplicative inverse a^{-1} with $a \cdot a^{-1} = 1$;
- multiplication distributes over addition: $a(b + c) = ab + ac$.

An **ordered field** is a field F together with a subset $F^+ \subseteq F$, the positive elements, such that:

- F^+ is closed under addition and under multiplication;
- for every $a \in F$, exactly one of the following holds: $a \in F^+$, $-a \in F^+$, $a = 0$.

We then write $a < b$ to mean $b - a \in F^+$.

If $A \subseteq F$, an **upper bound** for A is an element $b \in F$ with $a \leq b$ for every $a \in A$; we say A is bounded above if it has one. A **supremum** (or least upper bound) of A is an upper bound b such that $b \leq c$ for every upper bound c of A . Lower bounds and **infima** are defined the same way with the inequalities reversed.

A **complete ordered field** is an ordered field F in which every nonempty subset that is bounded above has a supremum in F .

1. Let X, Y, Z be sets. Prove the following, in each case by showing the two containments.
 - (a) $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$.
 - (b) $X \setminus (Y \cup Z) = (X \setminus Y) \cap (X \setminus Z)$.
 - (c) $Y \subseteq Z$ if and only if $Y \cup Z = Z$.
2. Let $S = \{x \in \mathbb{Q} \mid x > 0 \text{ and } x^2 < 3\}$, regarded as a subset of $\mathbb{Q}$.
 - (a) Show that S is nonempty and bounded above in $\mathbb{Q}$.
 - (b) Show that S has no supremum in $\mathbb{Q}$. You may use that $\sqrt{3}$ is irrational.
 - (c) What is $\sup S$ when S is regarded as a subset of $\mathbb{R}$? Which axiom guarantees it exists?
3. Let $f : [0, 1] \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} \frac{1}{\sqrt{x}} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Is $f(x)$ integrable? why or why not? *hint: how would you find a step function $t(x)$ with $t(x) \geq f(x)$?*

4. Suppose F is an ordered field.
 - (a) Show that if $0 < a < b$ in F , then $a^2 < b^2$.
 - (b) Suppose $0 < a < b$ in F and that there exist positive elements $\sqrt{a}, \sqrt{b} \in F$ with $\sqrt{a}^2 = a, \sqrt{b}^2 = b$. Show in this case that $\sqrt{a} < \sqrt{b}$.
5. Suppose F is an ordered field.

- (a) Show that if $c \in F$ and $c > 0$ then $c^{-1} > 0$, and that if $c < 0$ then $c^{-1} < 0$.
- (b) Suppose $a < b$ are nonzero elements of F which are either both positive or both negative. Show that $b^{-1} < a^{-1}$.
- (c) Does the conclusion of part (b) hold for every pair of nonzero elements $a < b$ of F ? Either prove that it does, or give a counterexample.
6. Use the definition of a limit to show the following:
- (a) $\lim_{x \rightarrow 2} x^2 = 4$
- (b) $\lim_{x \rightarrow 9} \sqrt{x} = 3$
7. Let $S = \{\sqrt{x} \mid 0 \leq x < 9\} \subseteq \mathbb{R}$.
- (a) Show that $\sup S = 3$.
- (b) Compare your argument with the one you gave for $\lim_{x \rightarrow 9} \sqrt{x} = 3$ in the previous problem. In part (a), what plays the role of ϵ , and what plays the role of δ ?
8. Let A and B be nonempty subsets of $\mathbb{R}$ which are bounded above.
- (a) Show that if $A \subseteq B$ then $\sup A \leq \sup B$.
- (b) Show that $\sup(A \cup B) = \max\{\sup A, \sup B\}$.
- (c) Suppose $A \cap B$ is nonempty. Is it always true that $\sup(A \cap B) = \min\{\sup A, \sup B\}$? Either prove it, or give an example showing it can fail.
9. Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and integrable, and let λ be any real number. Show that λf is integrable and

$$\int_a^b \lambda f(x) dx = \lambda \int_a^b f(x) dx.$$

there are two cases. For $\lambda > 0$, multiplying by λ preserves inequalities; for $\lambda < 0$ it reverses them, so step functions below f become step functions above λf , and $\sup$ and $\inf$ change places.

10. (a) Show, directly from the definition of the integral, that

$$\int_0^1 x dx = \frac{1}{2}.$$

hint: cut $[0, 1]$ into n equal pieces. Let s_n be the step function taking the value $\frac{k-1}{n}$ on $[\frac{k-1}{n}, \frac{k}{n})$, and let t_n take the value $\frac{k}{n}$ there. Check that $s_n(x) \leq x \leq t_n(x)$, compute $\int_0^1 s_n$ and $\int_0^1 t_n$, and then squeeze.

- (b) Show, directly from the definition of the integral, that

$$\int_0^1 x^2 dx = \frac{1}{3}.$$

hint: the same set-up, with s_n now taking the value $(\frac{k-1}{n})^2$ and t_n taking the value $(\frac{k}{n})^2$ on $[\frac{k-1}{n}, \frac{k}{n})$. You may use $1^2 + 2^2 + \cdots + m^2 = \frac{m(m+1)(2m+1)}{6}$.

11. Now the general case.

- (a) (Bernoulli's inequality) Show by induction that for every real number $x \geq -1$ and every integer $n \geq 1$,

$$(1+x)^n \geq 1+nx.$$

hint: in the inductive step you will want to multiply the inequality $(1+x)^n \geq 1+nx$ by $(1+x)$. Why is that allowed?

- (b) Deduce that $p+1 < 2^{p+1}$ for every integer $p \geq 1$.

- (c) Prove that $\sum_{k=1}^{n-1} k^p < \frac{n^{p+1}}{p+1}$ for every integer $n \geq 2$ and every integer $p \geq 1$, by induction on n . The base case $n = 2$ is exactly part (b); for the inductive step, expand $(n+1)^{p+1}$ using the binomial theorem.

12. Recall that a function $f : X \rightarrow Y$ is a subset $f \subseteq X \times Y$ such that for every $x \in X$ there is exactly one $y \in Y$ with $(x, y) \in f$.

For each of the following, decide whether it is a function from the first set to the second. Justify your answer — if it is not a function, say which part of the definition fails.

1. $\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid xy = 1\}$
2. $\{(x, y) \in (0, \infty) \times \mathbb{R} \mid xy = 1\}$
3. $\{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid 2y = x\}$
4. $\{(x, y) \in \mathbb{Z} \times \mathbb{Q} \mid 2y = x\}$
5. $\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y^3 = x\}$
6. $\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x^2 + y^2 = 1\}$

13. Recall the axioms for an area function. We have a collection $\mathcal{M}$ of subsets of $\mathbb{R}^2$, called measurable, and a function $a : \mathcal{M} \rightarrow \mathbb{R}$ such that

1. if $S, T \in \mathcal{M}$ then so are $S \cup T$, $S \cap T$, $S \setminus T$, and $a(S \cup T) = a(S) + a(T) - a(S \cap T)$,
 2. $a(S) \geq 0$ for every $S \in \mathcal{M}$,
 3. $a(\emptyset) = 0$,
 4. a is invariant under translation,
 5. the unit square has area 1.
- (a) Show that $a(S \cup T) \leq a(S) + a(T)$ for any $S, T \in \mathcal{M}$.

- (b) Show that if $S \subseteq T$ are both in $\mathcal{M}$, then $a(S) \leq a(T)$.

- (c) Show that a single point in the plane has area 0. (You may assume that points are measurable.)
hint: how many points can you fit inside the unit square?